🛠 **Bash Script – project_initializer.sh**

This Bash script automates the setup of a working environment for a technical team. It demonstrates my ability to organize a project structure, manage system users and groups, apply security permissions, and produce a packaged environment ready for deployment or archiving.

▫ **Detailed Steps of the Script**

1. ▫ **Welcome Message**

```bash
echo "Welcome $(whoami)!"
echo "$(date)"
```

Displays the name of the user running the script along with the date and time. This logs the execution for auditing.

2. **Creating Project Structure**

```
mkdir -p projects/users_team
cd projects
mkdir admin_team security_team
```

Creates the main `projects` directory with three subdirectories for teams: `users_team`, `admin_team`, and `security_team`.

3. **Adding Members to the "users" Team**

```
cd users_team
mkdir Gabens Don Danirode Lyncold
```

Creates subdirectories representing user folders within the team.

4. 🔙 **Returning to the Root Directory**

```
cd ../..
```

Returns to the top-level directory to continue system operations.

5. **Creating a System User and Group**

```
sudo useradd -m apm
sudo groupadd admin_team
sudo usermod -aG admin_team apm
sudo chgrp admin_team projects/admin_team
sudo passwd apm
```

Creates a user `apm` and a group `admin_team`. The user is added to the `admin_team` group, and the group is assigned to the `admin_team` folder. The password for `apm` is set for real use.

6. **Generating and Assigning a History File**

```
touch projects/users_team/history_command
history > projects/users_team/history_command
sudo chown apm projects/users_team/history_command
```

Logs the command history to a file `history_command`, assigns ownership to `apm`, simulating a traceability or audit system.

7. **Restricting Access**

```
sudo chmod o-r projects/admin_team projects/security_team
```

Revokes read permissions for other users on the sensitive `admin_team` and `security_team` directories, simulating access control by role.

8. 📦 **Archiving the Environment**

```
tar -czvf projects.tar.gz projects
```

Compresses the entire `projects` environment into a `.tar.gz` archive — useful for deliverables, backup, or deployment.

♻️ Cleanup Environment – remove_project_initializer.sh

A second script is provided to clean up everything created by the main script:

```
#!/bin/bash

sudo userdel apm
sudo groupdel admin_team
sudo rm -rf /home/apm projects projects.tar.gz
```

This script will:

- Delete the `apm` user
- Delete the `admin_team` group
- Remove all files and directories related to the project

💡 **This script is useful for:**

- Repeated testing
- Temporary environments
- Demonstrations

✅ Final Results

- A complete team-based environment (with users, permissions, and structure)
- A functional and secure system user
- An associated history file
- A `.tar.gz` archive of the project
- A professional cleanup script

Skills demonstrated:

- ✅ Linux system administration (users, groups, permissions)
- ✅ Automation via Bash scripting
- ✅ Security and access management
- ✅ Archiving and preparing deliverables

- ✅ Creating and cleaning up a reproducible environment